lambda-appl

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1 Code for "Plethysm for characters of relative operads and PROPs" – demonstration notebook

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This is the SageMath version of `mathematica/lambda-appl.nb`. Each code cell starts with a comment giving the Mathematica input it replaces. Most of the Mathematica package is built into SageMath, which is used directly; `relative_plethysm.py` only adds what SageMath does not provide (see the README).

[1]:
```
# Get["Lambda`"]
from relative_plethysm import *
from IPython.display import Markdown, display

def show_poly(f, sym='S'):
    display(Markdown(f"${print_poly(f, sym)}$"))
```

1.1 Partitions

The functions `tr`, `hook`, `\[ScriptZ]` and `\[Chi]` of the Mathematica code are the methods `conjugate`, `dimension` and `centralizer_size` of partitions, and the scalar product $\chi^\lambda(\mu) = \langle s_\lambda, p_\mu \rangle$.

[2]:
```
# tr[{5, 4, 4, 1}]
la = Partition([5, 4, 4, 1])
la.conjugate()
```

[2]:
```
[4, 3, 3, 3, 1]
```

[3]:
```
# hook[{5, 4, 4, 1}]
la.dimension()
```

[3]:
```
21021
```

[4]:
```
# \[ScriptZ][{5, 4, 4, 1}]
la.centralizer_size()
```

[4]:
```
160
```

[5]:
```
# \[Chi][{3, 2, 1}, {3, 3}]
s[3, 2, 1].scalar(p[3, 3])
```

[5]:
```
-2
```

1.2 Projections

`view[n][f]` is `f.expand(n)`, or `expand(f, n)` for symmetric functions in several alphabets.

[6]:
```
# TableForm[Comap[{Identity, view[3]}] /@ {m[{5, 4, 4, 2}, x], m[{1}, x],␣
 ↪m[{2}, x], m[{2, 1}, x], m[{1, 1, 1}, x]}, ...]
table([("f", "f in Λ_3")] + [(f, f.expand(3)) for f in [m[5, 4, 4, 2], m[1],␣
 ↪m[2], m[2, 1], m[1, 1, 1]]])
```

[6]:
```
  f               f in Λ_3
  m[5, 4, 4, 2]   0
  m[1]            x0 + x1 + x2
  m[2]            x0^2 + x1^2 + x2^2
  m[2, 1]         x0^2*x1 + x0*x1^2 + x0^2*x2 + x1^2*x2 + x0*x2^2 + x1*x2^2
  m[1, 1, 1]      x0*x1*x2
```

[7]:
```
# TableForm[Comap[{Identity, Expand @* view[3]}] /@ {e[1, x], e[2, x], e[3, x],␣
 ↪e[1, x] * e[2, y]}, ...]
table([("f", "f in Λ_3")] + [(f, expand(f, 3)) for f in [e[1], e[2], e[3],␣
 ↪e[1](X) * e[2](Y)]])
```

[7]:
```
  f               f in Λ_3
  e[1]            x0 + x1 + x2
  e[2]            x0*x1 + x0*x2 + x1*x2
  e[3]            x0*x1*x2
  s[1] # s[1, 1]  x0*y0*y1 + x1*y0*y1 + x2*y0*y1 + x0*y0*y2 + x1*y0*y2 +
x2*y0*y2 + x0*y1*y2 + x1*y1*y2 + x2*y1*y2
```

[8]:
```
# TableForm[Comap[{Identity, Expand @* view[3]}] /@ {h[1, x], h[2, x], h[3, x],␣
 ↪h[1, x] * h[2, y]}, ...]
table([("f", "f in Λ_3")] + [(f, expand(f, 3)) for f in [h[1], h[2], h[3],␣
 ↪h[1](X) * h[2](Y)]])
```

[8]:
```
  f           f in Λ_3
  h[1]        x0 + x1 + x2
  h[2]        x0^2 + x0*x1 + x1^2 + x0*x2 + x1*x2 + x2^2
  h[3]        x0^3 + x0^2*x1 + x0*x1^2 + x1^3 + x0^2*x2 + x0*x1*x2 + x1^2*x2 +
x0*x2^2 + x1*x2^2 + x2^3
```

```
  s[1] # s[2]   x0*y0^2 + x1*y0^2 + x2*y0^2 + x0*y0*y1 + x1*y0*y1 + x2*y0*y1 +
x0*y1^2 + x1*y1^2 + x2*y1^2 + x0*y0*y2 + x1*y0*y2 + x2*y0*y2 + x0*y1*y2 +
x1*y1*y2 + x2*y1*y2 + x0*y2^2 + x1*y2^2 + x2*y2^2
```

[9]:
```
# TableForm[Comap[{Identity, Expand @* view[3]}] /@ {p[0, x], p[1, x], p[2, x],␣
 ↪p[3, x], p[1, x] * p[2, y]}, ...]
table([("f", "f in Λ_3")] + [(f, expand(f, 3)) for f in [p[0], p[1], p[2],␣
 ↪p[3], p[1](X) * p[2](Y)]])
```

[9]:
```
  f                             f in Λ_3
  p[]                           1
  p[1]                          x0 + x1 + x2
  p[2]                          x0^2 + x1^2 + x2^2
  p[3]                          x0^3 + x1^3 + x2^3
  -s[1] # s[1, 1] + s[1] # s[2]   x0*y0^2 + x1*y0^2 + x2*y0^2 + x0*y1^2 +
x1*y1^2 + x2*y1^2 + x0*y2^2 + x1*y2^2 + x2*y2^2
```

[10]:
```
# Reduce[view[5][2 * h[2, x]] == view[5][p[1, x]^2 + p[2, x]]]
(2 * h[2]).expand(5) == (p[1]^2 + p[2]).expand(5)
```

[10]:
```
True
```

[11]:
```
# alt[{2}, x], alt[{2, 1}, x], alt[{3, 2, 1}, x]
def alt(alpha):
    n = len(alpha)
    R = PolynomialRing(QQ, [f'x{i}' for i in range(n)])
    return matrix(R, n, n, lambda i, j: R.gen(i)^alpha[j]).det()

alt([2]), alt([2, 1]), alt([3, 2, 1])
```

[11]:
```
(x0^2,
 x0^2*x1 - x0*x1^2,
 x0^3*x1^2*x2 - x0^2*x1^3*x2 - x0^3*x1*x2^2 + x0*x1^3*x2^2 + x0^2*x1*x2^3 -
x0*x1^2*x2^3)
```

[12]:
```
# TableForm[Comap[{Identity, Expand @* view[3]}] /@ {s[{1}, x], s[{2}, x],␣
 ↪s[{2, 1}, x]}, ...]
table([("f", "f in Λ_3")] + [(f, f.expand(3)) for f in [s[1], s[2], s[2, 1]]])
```

[12]:
```
  f         f in Λ_3
  s[1]      x0 + x1 + x2
  s[2]      x0^2 + x0*x1 + x1^2 + x0*x2 + x1*x2 + x2^2
  s[2, 1]   x0^2*x1 + x0*x1^2 + x0^2*x2 + 2*x0*x1*x2 + x1^2*x2 + x0*x2^2 +
x1*x2^2
```

1.2.1 Sanity check

Check that $s_\lambda = \det(e_{\lambda'_i - i + j})_{1 \le i,j \le \ell(\lambda')}$, as expected.

[13]:
```
# chk[\[Lambda]_List, n_Integer] := ...; chk[{2, 1}, 4], chk[{3, 2, 1}, 4],␣
↪chk[{4, 2, 1}, 5]
def chk(la, n):
    lt = Partition(la).conjugate()
    E = lambda k: e[k] if k >= 0 else e.zero()
    jt = matrix(e, len(lt), len(lt), lambda i, j: E(lt[i] - i + j)).det()
    return jt.expand(n) == s[la].expand(n)

chk([2, 1], 4), chk([3, 2, 1], 4), chk([4, 2, 1], 5)
```

[13]:
```
(True, True, True)
```

1.3 Conversions

conv[t][f] is t(f), or tensor([t, t])(f) in two alphabets.

[14]:
```
# TableForm[Comap[{Identity, conv[m]}] /@ {e[5, x], h[2, x], h[3, x], h[4, x],␣
↪p[5, x], e[2, x] * h[3, y]}]
table([(f, m(f)) for f in [e[5], h[2], h[3], h[4], p[5]]]
      + [(e[2](X) * h[3](Y), tensor([m, m])(e[2](X) * h[3](Y)))])
```

[14]:
```
  e[5]            m[1, 1, 1, 1, 1]
  h[2]            m[1, 1] + m[2]
  h[3]            m[1, 1, 1] + m[2, 1] + m[3]
  h[4]            m[1, 1, 1, 1] + m[2, 1, 1] + m[2, 2] + m[3, 1] + m[4]
  p[5]            m[5]
  s[1, 1] # s[3]  m[1, 1] # m[1, 1, 1] + m[1, 1] # m[2, 1] + m[1, 1] # m[3]
```

[15]:
```
# TableForm[Comap[{Identity, conv[e]}] /@ {h[5, x], p[5, x], s[{2, 1, 1}, x],␣
↪s[{1, 1, 1, 1, 1}, x]}]
table([(f, e(f)) for f in [h[5], p[5], s[2, 1, 1], s[1, 1, 1, 1, 1]]])
```

[15]:
```
  h[5]              e[1, 1, 1, 1, 1] - 4*e[2, 1, 1, 1] + 3*e[2, 2, 1] + 3*e[3,
1, 1] - 2*e[3, 2] - 2*e[4, 1] + e[5]
  p[5]              e[1, 1, 1, 1, 1] - 5*e[2, 1, 1, 1] + 5*e[2, 2, 1] + 5*e[3,
1, 1] - 5*e[3, 2] - 5*e[4, 1] + 5*e[5]
  s[2, 1, 1]        e[3, 1] - e[4]
  s[1, 1, 1, 1, 1]  e[5]
```

[16]:
```
# TableForm[Comap[{Identity, conv[h]}] /@ {e[2, x], e[3, x], e[4, x], p[2, x],␣
↪p[3, x], p[4, x], s[{2, 1, 1}, x]}]
table([(f, h(f)) for f in [e[2], e[3], e[4], p[2], p[3], p[4], s[2, 1, 1]]])
```

[16]:
```
  e[2]        h[1, 1] - h[2]
  e[3]        h[1, 1, 1] - 2*h[2, 1] + h[3]
  e[4]        h[1, 1, 1, 1] - 3*h[2, 1, 1] + h[2, 2] + 2*h[3, 1] - h[4]
  p[2]        -h[1, 1] + 2*h[2]
  p[3]        h[1, 1, 1] - 3*h[2, 1] + 3*h[3]
```

```
  p[4]          -h[1, 1, 1, 1] + 4*h[2, 1, 1] - 2*h[2, 2] - 4*h[3, 1] + 4*h[4]
  s[2, 1, 1]    h[2, 1, 1] - h[2, 2] - h[3, 1] + h[4]
```

[17]:
```
# TableForm[Comap[{Identity, conv[p]}] /@ {e[2, x], e[3, x], e[4, x], e[5, x],␣
 ↪h[2, x], h[3, x], h[4, x], h[5, x], s[{2, 1}, x]}]
table([(f, p(f)) for f in [e[2], e[3], e[4], e[5], h[2], h[3], h[4], h[5], s[2,␣
 ↪1]]])
```

[17]:
```
  e[2]      1/2*p[1, 1] - 1/2*p[2]
  e[3]      1/6*p[1, 1, 1] - 1/2*p[2, 1] + 1/3*p[3]
  e[4]      1/24*p[1, 1, 1, 1] - 1/4*p[2, 1, 1] + 1/8*p[2, 2] + 1/3*p[3, 1] -
1/4*p[4]
  e[5]      1/120*p[1, 1, 1, 1, 1] - 1/12*p[2, 1, 1, 1] + 1/8*p[2, 2, 1] +
1/6*p[3, 1, 1] - 1/6*p[3, 2] - 1/4*p[4, 1] + 1/5*p[5]
  h[2]      1/2*p[1, 1] + 1/2*p[2]
  h[3]      1/6*p[1, 1, 1] + 1/2*p[2, 1] + 1/3*p[3]
  h[4]      1/24*p[1, 1, 1, 1] + 1/4*p[2, 1, 1] + 1/8*p[2, 2] + 1/3*p[3, 1] +
1/4*p[4]
  h[5]      1/120*p[1, 1, 1, 1, 1] + 1/12*p[2, 1, 1, 1] + 1/8*p[2, 2, 1] +
1/6*p[3, 1, 1] + 1/6*p[3, 2] + 1/4*p[4, 1] + 1/5*p[5]
  s[2, 1]   1/3*p[1, 1, 1] - 1/3*p[3]
```

Sanity checks:

[18]:
```
# And @@ Flatten @ Table[Simplify[conv[t][conv[t2][t[10, x]]] == t[10, x]], {t,␣
 ↪{e, h, p}}, {t2, {e, h, p}}]
all(t(t2(t[10])) == t[10] for t in [e, h, p] for t2 in [e, h, p])
```

[18]:
```
True
```

[19]:
```
# Simplify[conv[p][Sum[hook[\[Lambda]] * s[\[Lambda], x], {\[Lambda],␣
 ↪IntegerPartitions[10]}]] == p[1, x]^10
p(sum(la.dimension() * s[la] for la in Partitions(10))) == p[1]^10
```

[19]:
```
True
```

[20]:
```
# With[{\[Mu] = {3, 2, 1}}, Simplify[view[6][p[\[Mu], x]] ==␣
 ↪view[6][Sum[\[Chi][\[Lambda], \[Mu]] * s[\[Lambda], x], {\[Lambda],␣
 ↪IntegerPartitions[Total[\[Mu]]]}]]]]
mu = Partition([3, 2, 1])
p[mu].expand(6) == sum(s[la].scalar(p[mu]) * s[la] for la in Partitions(mu.
 ↪size())).expand(6)
```

[20]:
```
True
```

1.4 Plethysm

`pl[f, g, x]` is SageMath's plethysm `f(g)`, also when `g` is a symmetric function in several alphabets. `pl[f, {g, h}, x, y]`, for `f` in two alphabets, is `plethysm(f, g, h)`.

[21]:
```
# pl[p[3, x], p[5, x], x]
p[3](p[5])
```

[21]:
```
p[15]
```

[22]:
```
# Expand[pl[e[2, x], h[2, x], x]]
p(e[2](h[2]))
```

[22]:
```
1/8*p[1, 1, 1, 1] + 1/4*p[2, 1, 1] - 1/8*p[2, 2] - 1/4*p[4]
```

[23]:
```
# view[6][e[2, x]] /. Thread[Table[Subscript[x, i], {i, 6}] -> List @@␣
 ↪view[3][h[2, x]]]
lhs = e[2].expand(6)(*h[2].expand(3).monomials())
lhs
```

[23]:
```
x0^3*x1 + x0^2*x1^2 + x0*x1^3 + x0^3*x2 + 2*x0^2*x1*x2 + 2*x0*x1^2*x2 + x1^3*x2
+ x0^2*x2^2 + 2*x0*x1*x2^2 + x1^2*x2^2 + x0*x2^3 + x1*x2^3
```

[24]:
```
# Expand[view[3][pl[e[2, x], h[2, x], x]]]
rhs = e[2](h[2]).expand(3)
rhs, rhs == lhs
```

[24]:
```
(x0^3*x1 + x0^2*x1^2 + x0*x1^3 + x0^3*x2 + 2*x0^2*x1*x2 + 2*x0*x1^2*x2 + x1^3*x2
+ x0^2*x2^2 + 2*x0*x1*x2^2 + x1^2*x2^2 + x0*x2^3 + x1*x2^3,
 True)
```

[25]:
```
# pl[p[3, x], {p[4, y], f}, x, y]
P2 = tensor([p, p])
P2(p[3](p[4](Y)))
```

[25]:
```
p[] # p[12]
```

[26]:
```
# pl[p[3, x], {p[4, x], h[2, y]}, x, y]
P2(p[3](p[4](X)))
```

[26]:
```
p[12] # p[]
```

[27]:
```
# pl[p[2, x] * p[3, y], {p[4, x], h[2, y]}, x, y]
P2(plethysm(p[2](X) * p[3](Y), p[4](X), h[2](Y)))
```

[27]:
```
1/2*p[8] # p[3, 3] + 1/2*p[8] # p[6]
```

The grading variable $\hbar$ is a variable of the base ring. By default, SageMath's plethysm treats the variables of the base ring as elements of degree one ($p_k \circ \hbar = \hbar^k$), as the Mathematica code does.

[28]:
```
# FullSimplify[pl[h[2, x] * e[3, y], {\[HBar]^2 * e[2, x] * e[1, y], \[HBar] *
 ↪h[2, y]}, x, y]]
R.<hbar> = QQ[]
SymR = SymmetricFunctions(R)
sR, hR, eR, pR = SymR.s(), SymR.h(), SymR.e(), SymR.p()
XR, YR = tensor([sR[1], sR[[]]]), tensor([sR[[]], sR[1]])
tensor([pR, pR])(plethysm(hR[2](XR) * eR[3](YR), hbar^2 * eR[2](XR) *
 ↪eR[1](YR), hbar * hR[2](YR)))
```

[28]:
```
1/384*hbar^7*p[1, 1, 1, 1] # p[1, 1, 1, 1, 1, 1, 1, 1] + 1/128*hbar^7*p[1, 1, 1,
1] # p[2, 1, 1, 1, 1, 1, 1] - 1/128*hbar^7*p[1, 1, 1, 1] # p[2, 2, 1, 1, 1, 1] -
5/384*hbar^7*p[1, 1, 1, 1] # p[2, 2, 2, 1, 1] + 1/48*hbar^7*p[1, 1, 1, 1] # p[3,
3, 1, 1] - 1/64*hbar^7*p[1, 1, 1, 1] # p[4, 1, 1, 1, 1] - 1/64*hbar^7*p[1, 1, 1,
1] # p[4, 2, 1, 1] + 1/48*hbar^7*p[1, 1, 1, 1] # p[6, 1, 1] - 1/192*hbar^7*p[2,
1, 1] # p[1, 1, 1, 1, 1, 1, 1, 1] - 1/64*hbar^7*p[2, 1, 1] # p[2, 1, 1, 1, 1, 1,
1] + 1/64*hbar^7*p[2, 1, 1] # p[2, 2, 1, 1, 1, 1] + 5/192*hbar^7*p[2, 1, 1] #
p[2, 2, 2, 1, 1] - 1/24*hbar^7*p[2, 1, 1] # p[3, 3, 1, 1] + 1/32*hbar^7*p[2, 1,
1] # p[4, 1, 1, 1, 1] + 1/32*hbar^7*p[2, 1, 1] # p[4, 2, 1, 1] -
1/24*hbar^7*p[2, 1, 1] # p[6, 1, 1] + 1/384*hbar^7*p[2, 2] # p[1, 1, 1, 1, 1, 1,
1, 1] + 5/384*hbar^7*p[2, 2] # p[2, 1, 1, 1, 1, 1, 1] + 1/128*hbar^7*p[2, 2] #
p[2, 2, 1, 1, 1, 1] - 11/384*hbar^7*p[2, 2] # p[2, 2, 2, 1, 1] -
5/192*hbar^7*p[2, 2] # p[2, 2, 2, 2] + 1/48*hbar^7*p[2, 2] # p[3, 3, 1, 1] +
1/24*hbar^7*p[2, 2] # p[3, 3, 2] - 1/64*hbar^7*p[2, 2] # p[4, 1, 1, 1, 1] -
3/64*hbar^7*p[2, 2] # p[4, 2, 1, 1] - 1/32*hbar^7*p[2, 2] # p[4, 2, 2] +
1/48*hbar^7*p[2, 2] # p[6, 1, 1] + 1/24*hbar^7*p[2, 2] # p[6, 2] -
1/192*hbar^7*p[4] # p[2, 1, 1, 1, 1, 1, 1] - 1/64*hbar^7*p[4] # p[2, 2, 1, 1, 1,
1] + 1/64*hbar^7*p[4] # p[2, 2, 2, 1, 1] + 5/192*hbar^7*p[4] # p[2, 2, 2, 2] -
1/24*hbar^7*p[4] # p[3, 3, 2] + 1/32*hbar^7*p[4] # p[4, 2, 1, 1] +
1/32*hbar^7*p[4] # p[4, 2, 2] - 1/24*hbar^7*p[4] # p[6, 2]
```

1.5 Orthogonality

Character of the regular representation, in two variables.

[29]:
```
# Table[reg2[n, x, y], {n, 4}] // Column
def reg(n):
    return sum(tensor([s[la], s[la]]) for la in Partitions(n))

def reg2(n):
    return sum(tensor([p[la], p[la]]) / la.centralizer_size() for la in
 ↪Partitions(n))

for n in range(1, 5):
    print(reg2(n))
```

```
p[1] # p[1]
1/2*p[1, 1] # p[1, 1] + 1/2*p[2] # p[2]
1/6*p[1, 1, 1] # p[1, 1, 1] + 1/2*p[2, 1] # p[2, 1] + 1/3*p[3] # p[3]
```

```
1/24*p[1, 1, 1, 1] # p[1, 1, 1, 1] + 1/4*p[2, 1, 1] # p[2, 1, 1] + 1/8*p[2, 2] #
p[2, 2] + 1/3*p[3, 1] # p[3, 1] + 1/4*p[4] # p[4]
```

[30]:
```
# Table[conv[p][reg[n, x, y]] == reg2[n, x, y], {n, 5}] // Reduce
all(P2(reg(n)) == reg2(n) for n in range(1, 6))
```

[30]: True

[31]:
```
# Table[conv[s][reg2[n, x, y]] == reg[n, x, y], {n, 5}] // Reduce
all(S2(reg2(n)) == reg(n) for n in range(1, 6))
```

[31]: True

1.5.1 Computation from 11/03/2025

`adj[f, g]` is `g.skew_by(f)`, or `skew_by(g, f)` in several alphabets.

[32]:
```
# pl[h[2, x], {p[1, x]^3 * p[1, y], 0}, x, y] * h[2, z]
X3, Y3, Z3 = [tensor([s[1] if i == j else s[[]] for i in range(3)]) for j in␣
↪range(3)]
P3 = tensor([p, p, p])
f = h[2](X3^3 * Y3) * h[2](Z3)
P3(f)
```

[32]: `1/4*p[1, 1, 1, 1, 1, 1] # p[1, 1] # p[1, 1] + 1/4*p[1, 1, 1, 1, 1, 1] # p[1, 1] # p[2] + 1/4*p[2, 2, 2] # p[2] # p[1, 1] + 1/4*p[2, 2, 2] # p[2] # p[2]`

[33]:
```
# reg[2, y, z]
h[2](Y3 * Z3)
```

[33]: `s[] # s[1, 1] # s[1, 1] + s[] # s[2] # s[2]`

[34]:
```
# adj[reg[2, y, z], %37] // Simplify
g = skew_by(f, h[2](Y3 * Z3))
P3(g)
```

[34]: `1/2*p[1, 1, 1, 1, 1, 1] # p[] # p[] + 1/2*p[2, 2, 2] # p[] # p[]`

[35]:
```
# % == pl[h[2, x], p[1, x]^3, x]
g == h[2](p[1]^3)(X3)
```

[35]: True

1.6 Sanity check

Check that $h_n(x) \circ h_1(x)h_1(y) = \sum_{\lambda \vdash n} s_\lambda(x)s_\lambda(y)$ in two different ways.

[36]:

```
# rules = ...; With[{n = 1}, Simplify[view[5][reg[n, x, y]] ==␣
 ↪view[Length[%42]][h[n, z]] /. %43]]
monomials = expand(X * Y, 5).monomials()
[expand(reg(n), 5) == h[n].expand(len(monomials))(*monomials) for n in range(1,␣
 ↪4)]
```

[36]:
```
[True, True, True]
```

With the actual bisymmetric plethysm (this is the Cauchy identity, also in SageMath's documentation):

[37]:
```
# Reduce[Table[pl[h[n, x], {h[1, x] * h[1, y], g}, x, y] == reg2[n, x, y], {n,␣
 ↪6}]]
all(h[n](X * Y) == S2(reg2(n)) for n in range(1, 7))
```

[37]:
```
True
```

1.7 Computations

1.7.1 Stable twisted cohomology of automorphism groups of free groups

We have: $\mathcal{Q}(q,p) = \mathrm{triv}_p$ if $q \in \{0,1\}$ and $p \geq 1$, and 0 otherwise. Moreover $\mathcal{H} = \omega(\mathcal{P}) = \omega(\mathcal{F}(S(\mathcal{Q})))$ is the involution applied to the saturation of $\mathcal{Q}$. Unlike the Albanese cohomology, this is infinite dimensional in any fixed degree (because $\mathcal{Q}(1,0)$ contains an element of degree 0), so we can only compute the truncation up to some arity.

[38]:
```
# TableForm[Table[{chQ[d, x, y]}, {d, 0, 6}], ...]
EH = tensor([e, h])
table([("d", "ch(Q) in degree d")] + [(d, EH(chQ(d))) for d in range(7)])
```

[38]:

d	ch(Q) in degree d
0	e[1] # h[1]
1	e[] # h[1] + e[1] # h[2]
2	e[] # h[2] + e[1] # h[3]
3	e[] # h[3] + e[1] # h[4]
4	e[] # h[4] + e[1] # h[5]
5	e[] # h[5] + e[1] # h[6]
6	e[] # h[6] + e[1] # h[7]

[39]:
```
# chH_upto[2, 5, x, y] // printPoly[S, x, y]
show_poly(chH_upto(2, 5))
```

$S_{3,41}^{\oplus 3} \oplus S_{3,32} \oplus S_{3,31^2}^{\oplus 6} \oplus S_{3,2^21}^{\oplus 2} \oplus S_{3,21^3}^{\oplus 3} \oplus S_{21,41} \oplus S_{21,32}^{\oplus 5} \oplus S_{21,31^2}^{\oplus 7} \oplus S_{21,2^21}^{\oplus 8} \oplus S_{21,21^3}^{\oplus 9} \oplus S_{21,1^5}^{\oplus 3} \oplus S_{1^3,32}^{\oplus 2} \oplus S_{1^3,31^2} \oplus S_{1^3,2^21}^{\oplus 6} \oplus S_{1^3,21^3}^{\oplus 6} \oplus S_{1^3,1^5}^{\oplus 5} \oplus S_{2,31}^{\oplus 3} \oplus S_{2,2^2} \oplus S_{2,21^2}^{\oplus 6} \oplus S_{2,1^4}^{\oplus 2} \oplus S_{1^2,31} \oplus S_{1^2,2^2}^{\oplus 4} \oplus S_{1^2,21^2}^{\oplus 5} \oplus S_{1^2,1^4}^{\oplus 5} \oplus S_{1,21}^{\oplus 3} \oplus S_{1,1^3}^{\oplus 4} \oplus S_{\emptyset,1^2}^{\oplus 2}$

The characters of $\mathcal{P}$ and $\mathcal{H}$ in degree d, output arity m, input arity n (nonzero entries only). In the Mathematica version, the term $S_{\emptyset,1^2}^{\oplus 2}$ also appeared for $n = 2$ and every $m \geq 1$, because the truncation in `chP[d, m, n]` did not remove the terms of output arity 0.

[40]:
```
# TableForm[Map[printPoly[S, x, y], Table[chH[2, m, n, x, y], {m, 0, 5}, {n, 0,␣
↪5}], {2}], ...]
display(Markdown("\n".join(f"- $m = {m}$, $n = {n}$: ${print_poly(chH(2, m,␣
↪n))}$"
                           for m in range(6) for n in range(6) if chH(2, m,␣
↪n))))
```

- $m = 0$, $n = 2$: $S_{\emptyset,1^2}^{\oplus 2}$
- $m = 1$, $n = 3$: $S_{1,21}^{\oplus 3} \oplus S_{1,1^3}^{\oplus 4}$
- $m = 2$, $n = 4$: $S_{2,31}^{\oplus 3} \oplus S_{2,2^2} \oplus S_{2,21^2}^{\oplus 6} \oplus S_{2,1^4}^{\oplus 2} \oplus S_{1^2,31} \oplus S_{1^2,2^2}^{\oplus 4} \oplus S_{1^2,21^2}^{\oplus 5} \oplus S_{1^2,1^4}^{\oplus 5}$
- $m = 3$, $n = 5$: $S_{3,41}^{\oplus 3} \oplus S_{3,32} \oplus S_{3,31^2}^{\oplus 6} \oplus S_{3,2^21}^{\oplus 2} \oplus S_{3,21^3}^{\oplus 3} \oplus S_{21,41} \oplus S_{21,32}^{\oplus 5} \oplus S_{21,31^2}^{\oplus 7} \oplus S_{21,2^21}^{\oplus 8} \oplus S_{21,21^3}^{\oplus 9} \oplus S_{21,1^5}^{\oplus 3} \oplus S_{1^3,32}^{\oplus 2} \oplus S_{1^3,31^2} \oplus S_{1^3,2^21}^{\oplus 6} \oplus S_{1^3,21^3}^{\oplus 6} \oplus S_{1^3,1^5}^{\oplus 5}$

1.7.2 Albanese cohomology for IA_∞

We have: $\mathcal{Q}'(q,p) = \mathrm{triv}_p[p]$ if $q = 0$ and $p \geq 1$, $\mathrm{triv}_p[p-1]$ if $q = 1$ and $p \geq 2$, and 0 otherwise. Moreover $\mathcal{H}' = \omega(\mathcal{P}') = \omega(S(\mathcal{Q}'))$ is the involution applied to the saturation of $\mathcal{Q}'$.

The following function gives the character of $\mathcal{Q}'$ in homological degree d:

[41]:
```
# TableForm[Table[{chQ_prime[d, x, y]}, {d, 0, 6}], ...]
H2 = tensor([h, h])
table([("d", "ch(Q') in degree d")] + [(d, H2(chQ_prime(d))) for d in range(7)])
```

[41]:

d	ch(Q') in degree d
0	0
1	h[] # h[1] + h[1] # h[2]
2	h[] # h[2] + h[1] # h[3]
3	h[] # h[3] + h[1] # h[4]
4	h[] # h[4] + h[1] # h[5]
5	h[] # h[5] + h[1] # h[6]
6	h[] # h[6] + h[1] # h[7]

Now, the saturation $\mathcal{P}' = S(\mathcal{Q}')$ has character $\sum_{n\geq 1} h_n(x) \circ (\mathrm{ch}(\mathcal{Q}'), 0)$. The following function gives its part in degree exactly d (without the $(-\hbar)^d$).

[42]:
```
# TableForm[Table[{d, printPoly[S, x, y][chP_prime[d, x, y]]}, {d, 3}], ...]
display(Markdown("\n".join(f"- $d = {d}$: ${print_poly(chP_prime(d))}$" for d␣
↪in range(1, 4))))
```

- $d = 1$: $S_{1,2} \oplus S_{\emptyset,1}$
- $d = 2$: $S_{2,4} \oplus S_{2,2^2} \oplus S_{1^2,31} \oplus S_{1,3}^{\oplus 2} \oplus S_{1,21} \oplus S_{\emptyset,2}^{\oplus 2}$
- $d = 3$: $S_{3,6} \oplus S_{3,42} \oplus S_{3,2^3} \oplus S_{21,51} \oplus S_{21,42} \oplus S_{21,321} \oplus S_{1^3,41^2} \oplus S_{1^3,3^2} \oplus S_{2,5}^{\oplus 2} \oplus S_{2,41}^{\oplus 2} \oplus S_{2,32}^{\oplus 2} \oplus S_{2,2^21} \oplus S_{1^2,5} \oplus S_{1^2,41}^{\oplus 2} \oplus S_{1^2,32}^{\oplus 2} \oplus S_{1^2,31^2} \oplus S_{1,4}^{\oplus 4} \oplus S_{1,31}^{\oplus 3} \oplus S_{1,2^2}^{\oplus 2} \oplus S_{\emptyset,3}^{\oplus 3} \oplus S_{\emptyset,21}$

1.7.3 Calculations

Now we can find the decomposition of $\mathrm{ch}(\mathcal{H}')$ in some degree. The file `chH_prime.txt` contains $\mathrm{ch}(\mathcal{H}'_d)$ for $d \leq 10$; it was produced by `compute_chH_prime.py`.

[43]:
```
# Get[FileNameJoin[NotebookDirectory[], "chH_prime.mx"]]
chH_prime_table = load_chH_prime("chH_prime.txt")
```

[44]:
```
# printPoly[V, x, y][chH_prime[1, x, y]]
show_poly(chH_prime_table[1], 'V')
```

$V_{1,1^2} \oplus V_{\emptyset,1}$

[45]:
```
# printPoly[V, x, y][chH_prime[2, x, y]]
show_poly(chH_prime_table[2], 'V')
```

$V_{2,21^2} \oplus V_{1^2,2^2} \oplus V_{1^2,1^4} \oplus V_{1,21} \oplus V_{1,1^3}^{\oplus 2} \oplus V_{\emptyset,1^2}^{\oplus 2}$

[46]:
```
# printPoly[V, x, y][chH_prime[3, x, y]]
show_poly(chH_prime_table[3], 'V')
```

$V_{3,31^3} \oplus V_{3,2^3} \oplus V_{21,321} \oplus V_{21,2^21^2} \oplus V_{21,21^4} \oplus V_{1^3,3^2} \oplus V_{1^3,2^21^2} \oplus V_{1^3,1^6} \oplus V_{2,31^2} \oplus V_{2,2^21}^{\oplus 2} \oplus V_{2,21^3}^{\oplus 2} \oplus V_{2,1^5} \oplus V_{1^2,32} \oplus V_{1^2,2^21}^{\oplus 2} \oplus V_{1^2,21^3}^{\oplus 2} \oplus V_{1^2,1^5}^{\oplus 2} \oplus V_{1,2^2}^{\oplus 2} \oplus V_{1,21^2}^{\oplus 3} \oplus V_{1,1^4}^{\oplus 4} \oplus V_{\emptyset,21} \oplus V_{\emptyset,1^3}^{\oplus 3}$

[47]:
```
# printPoly[V, x, y][chH_prime[4, x, y]]
show_poly(chH_prime_table[4], 'V')
```

$V_{4,41^4} \oplus V_{4,32^21} \oplus V_{31,421^2} \oplus V_{31,3^22} \oplus V_{31,32^21} \oplus V_{31,321^3} \oplus V_{31,31^5} \oplus V_{31,2^31^2} \oplus V_{2^2,42^2} \oplus V_{2^2,3^21^2} \oplus V_{2^2,321^3} \oplus V_{2^2,2^4} \oplus V_{2^2,2^21^4} \oplus V_{21^2,431} \oplus V_{21^2,3^21^2} \oplus V_{21^2,32^21} \oplus V_{21^2,321^3} \oplus V_{21^2,2^31^2} \oplus V_{21^2,2^21^4} \oplus V_{21^2,21^6} \oplus V_{1^4,4^2} \oplus V_{1^4,3^21^2} \oplus V_{1^4,2^4} \oplus V_{1^4,2^21^4} \oplus V_{1^4,1^8} \oplus V_{3,41^3} \oplus V_{3,32^2}^{\oplus 2} \oplus V_{3,321^2}^{\oplus 2} \oplus V_{3,31^4}^{\oplus 2} \oplus V_{3,2^31}^{\oplus 2} \oplus V_{3,2^21^3} \oplus V_{3,21^5} \oplus V_{21,421} \oplus V_{21,3^21}^{\oplus 2} \oplus V_{21,32^2}^{\oplus 2} \oplus V_{21,321^2}^{\oplus 4} \oplus V_{21,31^4}^{\oplus 2} \oplus V_{21,2^31}^{\oplus 3} \oplus V_{21,2^21^3}^{\oplus 5} \oplus V_{21,21^5}^{\oplus 3} \oplus V_{21,1^7} \oplus V_{1^3,43} \oplus V_{1^3,3^21}^{\oplus 2} \oplus V_{1^3,321^2}^{\oplus 2} \oplus V_{1^3,2^31}^{\oplus 2} \oplus V_{1^3,2^21^3}^{\oplus 3} \oplus V_{1^3,21^5}^{\oplus 2} \oplus V_{1^3,1^7}^{\oplus 2} \oplus V_{2,321}^{\oplus 3} \oplus V_{2,31^3}^{\oplus 3} \oplus V_{2,2^3}^{\oplus 4} \oplus V_{2,2^21^2}^{\oplus 5} \oplus V_{2,21^4}^{\oplus 6} \oplus V_{2,1^6}^{\oplus 2} \oplus V_{1^2,3^2}^{\oplus 2} \oplus V_{1^2,321}^{\oplus 3} \oplus V_{1^2,31^3} \oplus V_{1^2,2^3} \oplus V_{1^2,2^21^2}^{\oplus 8} \oplus V_{1^2,21^4}^{\oplus 5} \oplus V_{1^2,1^6}^{\oplus 5} \oplus V_{1,32} \oplus V_{1,31^2} \oplus V_{1,2^21}^{\oplus 6} \oplus V_{1,21^3}^{\oplus 7} \oplus V_{1,1^5}^{\oplus 7} \oplus V_{\emptyset,2^2}^{\oplus 2} \oplus V_{\emptyset,21^2}^{\oplus 2} \oplus V_{\emptyset,1^4}^{\oplus 5}$

[48]:
```
# printPoly[V, x, y][chH_prime[5, x, y]]
show_poly(chH_prime_table[5], 'V')
```

$V_{5,51^5} \oplus V_{5,42^21^2} \oplus V_{5,3^22^2} \oplus V_{41,521^3} \oplus V_{41,4321} \oplus V_{41,42^3} \oplus V_{41,42^21^2} \oplus V_{41,421^4} \oplus V_{41,41^6} \oplus V_{41,3^31} \oplus V_{41,3^21^2} \oplus V_{41,32^31} \oplus V_{41,32^21^3} \oplus V_{32,52^21} \oplus V_{32,43^2} \oplus V_{32,4321} \oplus V_{32,431^3} \oplus V_{32,42^21^2} \oplus V_{32,421^4} \oplus V_{32,3^22^2} \oplus V_{32,3^21^4} \oplus V_{32,32^31} \oplus V_{32,32^21^3} \oplus V_{32,321^5} \oplus V_{32,2^41^2} \oplus V_{31^2,531^2} \oplus V_{31^2,4^22} \oplus V_{31^2,4321} \oplus V_{31^2,431^3} \oplus V_{31^2,42^3} \oplus V_{31^2,42^21^2} \oplus V_{31^2,421^4} \oplus V_{31^2,3^31} \oplus V_{31^2,3^21^2}^{\oplus 2} \oplus V_{31^2,32^31} \oplus V_{31^2,32^21^3}^{\oplus 2} \oplus V_{31^2,321^5} \oplus V_{31^2,31^7} \oplus V_{31^2,2^5} \oplus V_{31^2,2^31^4} \oplus V_{2^21,532} \oplus V_{2^21,4^21^2} \oplus V_{2^21,4321} \oplus V_{2^21,431^3} \oplus V_{2^21,42^21^2} \oplus V_{2^21,3^22^2} \oplus V_{2^21,3^21^2} \oplus V_{2^21,3^21^4}^{\oplus 2} \oplus V_{2^21,32^31} \oplus V_{2^21,32^21^3} \oplus V_{2^21,321^5} \oplus V_{2^21,2^41^2} \oplus V_{2^21,2^31^4} \oplus V_{2^21,2^21^6} \oplus V_{21^3,541} \oplus V_{21^3,4^21^2} \oplus V_{21^3,4321} \oplus V_{21^3,431^3} \oplus V_{21^3,3^22^2} \oplus V_{21^3,3^21^2} \oplus V_{21^3,3^21^4} \oplus V_{21^3,32^31} \oplus V_{21^3,32^21^3} \oplus V_{21^3,321^5} \oplus V_{21^3,2^41^2} \oplus V_{21^3,2^31^4} \oplus V_{21^3,2^21^6} \oplus V_{21^3,21^8} \oplus V_{1^5,5^2} \oplus V_{1^5,4^21^2} \oplus V_{1^5,3^22^2} \oplus V_{1^5,3^21^4} \oplus V_{1^5,2^41^2} \oplus V_{1^5,2^21^6} \oplus V_{1^5,1^{10}} \oplus V_{4,51^4} \oplus V_{4,42^21}^{\oplus 2} \oplus V_{4,421^3}^{\oplus 2} \oplus V_{4,41^5}^{\oplus 2} \oplus V_{4,3^3} \oplus V_{4,3^21}^{\oplus 2} \oplus V_{4,32^3}^{\oplus 2} \oplus V_{4,32^21^2}^{\oplus 3} \oplus V_{4,321^4} \oplus V_{4,31^6} \oplus V_{4,2^31^3} \oplus V_{31,521^2} \oplus V_{31,432}^{\oplus 2} \oplus V_{31,431^2}^{\oplus 2} \oplus V_{31,42^21}^{\oplus 4} \oplus V_{31,421^3}^{\oplus 4} \oplus V_{31,41^5}^{\oplus 2} \oplus V_{31,3^3}^{\oplus 2} \oplus V_{31,3^21}^{\oplus 5} \oplus V_{31,3^21^3}^{\oplus 3} \oplus V_{31,32^3}^{\oplus 3} \oplus V_{31,32^21^2}^{\oplus 8} \oplus V_{31,321^4}^{\oplus 6} \oplus V_{31,31^6}^{\oplus 3} \oplus V_{31,2^41}^{\oplus 3} \oplus V_{31,2^31^3}^{\oplus 4} \oplus V_{31,2^21^5}^{\oplus 2} \oplus V_{31,21^7} \oplus V_{2^2,52^2} \oplus V_{2^2,432}^{\oplus 2} \oplus V_{2^2,431^2}^{\oplus 2} \oplus V_{2^2,42^21}^{\oplus 2} \oplus V_{2^2,421^3}^{\oplus 2} \oplus V_{2^2,3^21}^{\oplus 3} \oplus V_{2^2,3^21^3}^{\oplus 4} \oplus V_{2^2,32^3}^{\oplus 2} \oplus V_{2^2,32^21^2}^{\oplus 4} \oplus V_{2^2,321^4}^{\oplus 5} \oplus V_{2^2,31^6}^{\oplus 2} \oplus V_{2^2,2^41}^{\oplus 3} \oplus V_{2^2,2^31^3}^{\oplus 3} \oplus V_{2^2,2^21^5}^{\oplus 3} \oplus V_{2^2,21^7} \oplus V_{21^2,531} \oplus V_{21^2,4^21}^{\oplus 2} \oplus V_{21^2,432}^{\oplus 2} \oplus V_{21^2,431^2}^{\oplus 4} \oplus V_{21^2,42^21}^{\oplus 2} \oplus V_{21^2,421^3}^{\oplus 2} \oplus V_{21^2,3^21}^{\oplus 5} \oplus V_{21^2,3^21^3}^{\oplus 6} \oplus V_{21^2,32^3}^{\oplus 3} \oplus V_{21^2,32^21^2}^{\oplus 7} \oplus V_{21^2,321^4}^{\oplus 6} \oplus V_{21^2,31^6}^{\oplus 2} \oplus V_{21^2,2^41}^{\oplus 4} \oplus V_{21^2,2^31^3}^{\oplus 6} \oplus V_{21^2,2^21^5}^{\oplus 6} \oplus V_{21^2,21^7}^{\oplus 3} \oplus V_{21^2,1^9} \oplus V_{1^4,54} \oplus V_{1^4,4^21}^{\oplus 2} \oplus V_{1^4,431^2}^{\oplus 2} \oplus V_{1^4,3^21}^{\oplus 3} \oplus V_{1^4,3^21^3}^{\oplus 2} \oplus V_{1^4,32^3} \oplus V_{1^4,32^21^2}^{\oplus 2} \oplus V_{1^4,321^4}^{\oplus 2} \oplus V_{1^4,2^41}^{\oplus 2} \oplus V_{1^4,2^31^3}^{\oplus 3} \oplus V_{1^4,2^21^5}^{\oplus 3} \oplus V_{1^4,21^7}^{\oplus 2} \oplus$

$$V_{1^4,1^9}^{\oplus 2} \oplus V_{3,42^2} \oplus V_{3,421^2}^{\oplus 3} \oplus V_{3,41^4}^{\oplus 3} \oplus V_{3,3^2 2}^{\oplus 4} \oplus V_{3,3^2 1^2} \oplus V_{3,32^2 1}^{\oplus 9} \oplus V_{3,321^3}^{\oplus 7} \oplus V_{3,31^5}^{\oplus 6} \oplus V_{3,2^4} \oplus V_{3,2^3 1^2}^{\oplus 7} \oplus V_{3,2^2 1^4}^{\oplus 4} \oplus V_{3,21^6}^{\oplus 3} \oplus V_{21,431}^{\oplus 3} \oplus V_{21,42^2}^{\oplus 3} \oplus V_{21,421^2}^{\oplus 4} \oplus V_{21,41^4} \oplus V_{21,3^2 2}^{\oplus 5} \oplus V_{21,3^2 1^2}^{\oplus 9} \oplus V_{21,32^2 1}^{\oplus 12} \oplus V_{21,321^3}^{\oplus 16} \oplus V_{21,31^5}^{\oplus 7} \oplus V_{21,2^4}^{\oplus 6} \oplus V_{21,2^3 1^2}^{\oplus 14} \oplus V_{21,2^2 1^4}^{\oplus 15} \oplus V_{21,21^6}^{\oplus 9} \oplus V_{21,1^8}^{\oplus 3} \oplus V_{1^3,4^2}^{\oplus 2} \oplus V_{1^3,431}^{\oplus 3} \oplus V_{1^3,421^2} \oplus V_{1^3,3^2 2} \oplus V_{1^3,3^2 1^2}^{\oplus 8} \oplus V_{1^3,32^2 1}^{\oplus 5} \oplus V_{1^3,321^3}^{\oplus 7} \oplus V_{1^3,31^5} \oplus V_{1^3,2^4}^{\oplus 5} \oplus V_{1^3,2^3 1^2}^{\oplus 7} \oplus V_{1^3,2^2 1^4}^{\oplus 11} \oplus V_{1^3,21^6}^{\oplus 6} \oplus V_{1^3,1^8}^{\oplus 5} \oplus V_{2,421} \oplus V_{2,41^3} \oplus V_{2,3^2 1}^{\oplus 3} \oplus V_{2,32^2}^{\oplus 7} \oplus V_{2,321^2}^{\oplus 10} \oplus V_{2,31^4}^{\oplus 8} \oplus V_{2,2^3 1}^{\oplus 11} \oplus V_{2,2^2 1^3}^{\oplus 15} \oplus V_{2,21^5}^{\oplus 12} \oplus V_{2,1^7}^{\oplus 5} \oplus V_{1^2,43} \oplus V_{1^2,421} \oplus V_{1^2,3^2 1}^{\oplus 6} \oplus V_{1^2,32^2}^{\oplus 3} \oplus V_{1^2,321^2}^{\oplus 11} \oplus V_{1^2,31^4}^{\oplus 4} \oplus V_{1^2,2^3 1}^{\oplus 11} \oplus V_{1^2,2^2 1^3}^{\oplus 18} \oplus V_{1^2,21^5}^{\oplus 13} \oplus V_{1^2,1^7}^{\oplus 9} \oplus V_{1,3^2}^{\oplus 2} \oplus V_{1,321}^{\oplus 5} \oplus V_{1,31^3}^{\oplus 3} \oplus V_{1,2^3}^{\oplus 5} \oplus V_{1,2^2 1^2}^{\oplus 15} \oplus V_{1,21^4}^{\oplus 14} \oplus V_{1,1^6}^{\oplus 12} \oplus V_{\emptyset,32} \oplus V_{\emptyset,2^2 1}^{\oplus 4} \oplus V_{\emptyset,21^3}^{\oplus 5} \oplus V_{\emptyset,1^5}^{\oplus 7}$$

[49]:
```
# printPoly[V, x, y][chH_prime[2, x, y]] // TeXForm // CopyToClipboard
print(print_poly(chH_prime_table[2], 'V'))
```

```
V_{2,21^{2}} \oplus V_{1^{2},2^{2}} \oplus V_{1^{2},1^{4}} \oplus V_{1,21}
\oplus V_{1,1^{3}}^{\oplus 2} \oplus V_{\varnothing,1^{2}}^{\oplus 2}
```

[50]:
```
# Table[chH_prime[d, x, y] /. s[\[Lambda]_, _] :> hook[\[Lambda]], {d, 1, 7}]
[sum(c * la.dimension() * mu.dimension() for (la, mu), c in chH_prime_table[d])␣
 ↪for d in range(1, 8)]
```

[50]:
```
[2, 12, 162, 4221, 182106, 11705807, 1046227328]
```

[51]:
```
# TeXForm[Transpose[Table[{d, Length[Expand[chH_prime[d, x, y]]], chH_prime[d,␣
 ↪x, y] /. _s -> 1}, {d, 1, 9}]]]
print(latex(matrix([[d, len(chH_prime_table[d]), sum(chH_prime_table[d].
 ↪coefficients())] for d in range(1, 10)]).transpose()))
```

```
\left(\begin{array}{rrrrrrrrr}
1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\
2 & 6 & 21 & 69 & 219 & 663 & 1915 & 5182 & 13330 \\
2 & 8 & 34 & 152 & 720 & 3634 & 19266 & 107018 & 619606
\end{array}\right)
```

[52]:
```
# Table[Length[Expand[chH_prime[d, x, y]]], {d, 1, 10}]
[len(chH_prime_table[d]) for d in range(1, 11)]
```

[52]:
```
[2, 6, 21, 69, 219, 663, 1915, 5182, 13330, 32876]
```

The stored characters agree with a fresh computation:

[53]:
```
all(chH_prime_table[d] == chH_prime(d) for d in range(1, 7))
```

[53]:
```
True
```